

MFGE 436/536 Final Design Report

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03/16/2026

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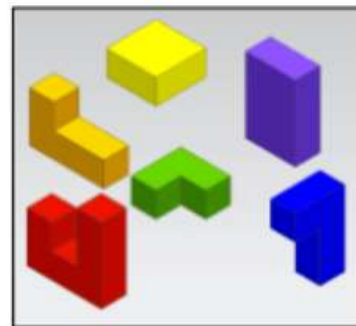
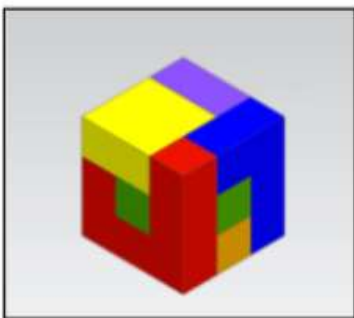
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Introduction

Project Overview

The team was contracted by LSME Inc to improve the manufacturing process of their puzzle cube. Overall the team was asked to double to production capacity of the line. LSME's starting process creates six interlocking pieces designed to fit together. The puzzle cube is produced using epoxy resin casting with small batch production. The workflow begins by measuring components A and B of the resin into equal quantities. It is then mixed and poured into molds to cure. After curing, the parts are demolded and deburred, then undergo quality assurance before they are assembled into the complete puzzle. This method allows for flexibility and prototyping but is limited by scale.



This manufacturing process is somewhat uncommon for this application; more common methods are used for products like the Rubik's cube. The Rubik's cube is created using injection molding, which consists of using pressure to force molten thermoplastics into a precise mold. After removal from the mold, the pieces undergo trimming and finishing to remove excess material, then followed by assembly. Throughout this process, automated quality control systems inspect

accuracy, color placement, and function. This standardized process helps manufacturers meet global demand while maintaining performance [1].

Design Methodology

The mission for this team has been to improve and expedite the manufacturing process. The method being used to ensure quality and identify opportunities for improvement in this project is the DMAIC Methodology. **Define, Measure, Analyze, Improve, and Control (DMAIC)** is a methodology commonly applied to both laboratory and commercial settings. As stated by ASQ, “DMAIC is a structured problem-solving approach used to improve existing processes that don’t meet performance standards or customer expectations” [2]. The methodology involves identifying the problem, quantifying the problem through data, analyzing the root cause, implementing solutions, and establishing controls to sustain improvements. The team applied every step of the DMAIC process in order to reach production goals.

Define Phase

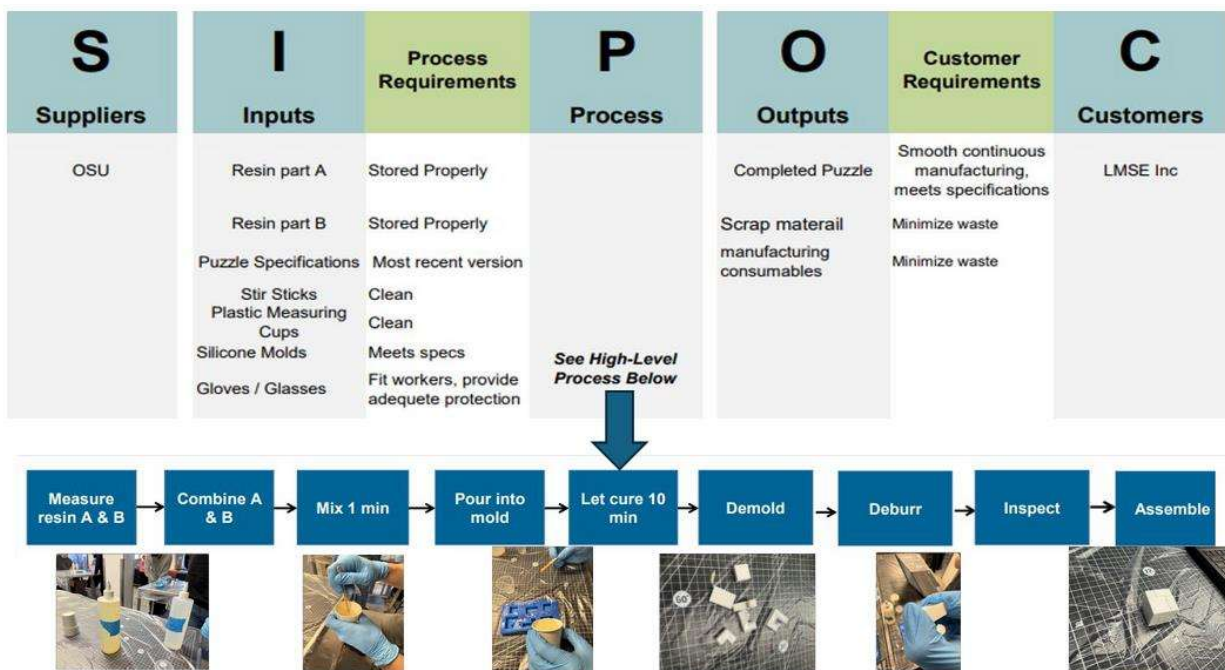
The first phase of the DMAIC methodology is Define. The purpose of the Define phase is to clearly define the problem and start to identify opportunities for improvement. Within this phase, a scope is clearly defined to align with the project team and stakeholders before data collection and analysis begins. This creates the foundation for the rest of the process and ensures everyone is on the same page.

Many Lean tools are used to help organizations define their current state. These tools include project charters, process flow diagrams, and SIPOC diagrams [3]. According to ASQ, another common tool used to help identify problems is Voice of the Customer (VoC), which includes surveys, questionnaires, and many more [2].

The project charter is a foundational document that clearly defines scope, boundaries, timeline, stakeholders, and responsibilities. Though this is not helpful in identifying problems, it facilitates communication once a problem is identified. This prevents ambiguity, scope creep, and misunderstandings as the project progresses. Our initial project charter can be seen below.

Project Charter																	
Problem Statement		Business Case & Benefits															
LMSE Inc. needs to meet increased demand for their puzzle cubes, and currently lacks manufacturing capacity. 150,000 puzzles must be delivered smoothly and continuously over the next year while minimizing waste and reducing takt time.		Increasing the manufacturing capacity of the company will increase the puzzles that can be sold. Reducing waste and adding continuous improvement to the process can improve every step of manufacturing, saving time and money. Without improvement to the current process LMSE will not be able to meet demand.															
Goal Statement		Timeline															
Increase LSMEs prodction capability from 75,000 to 150,000 units by the end of the year, and continuously improve the manufacturing process using LEAN principles. Hit a final takt time of 1.33 puzzles per minute.		<table><tr><th>Phase</th><th>Planned Completion Date</th><th>Actual</th></tr><tr><td>Define:</td><td>16-Jan</td><td rowspan="5">16-Jan</td></tr><tr><td>Measure:</td><td>6-Feb</td></tr><tr><td>Analyze:</td><td>6-Feb</td></tr><tr><td>Improve:</td><td>20-Feb</td></tr><tr><td>Control:</td><td>13-Mar</td></tr></table>		Phase	Planned Completion Date	Actual	Define:	16-Jan	16-Jan	Measure:	6-Feb	Analyze:	6-Feb	Improve:	20-Feb	Control:	13-Mar
Phase	Planned Completion Date	Actual															
Define:	16-Jan	16-Jan															
Measure:	6-Feb																
Analyze:	6-Feb																
Improve:	20-Feb																
Control:	13-Mar																
Scope - First/Last and In/Out		Team Members															
<u>1st Process Step</u>	Shaking and measuring the epoxy A and B	<u>Position</u>	<u>Person</u>	<u>Title</u>	<u>% of Time</u>												
<u>Last Process Step</u>	Inspecting / Assembly	Team Lead	Ross Hodson	Head Engineer	20%												
<u>In Scope:</u>	Epoxy casting, deburring, inspection, assembly	Team Member	Luncheng Zhang	Engineer	10%												
<u>Out of Scope:</u>	Distribution and Material Procurement	Team Member	Nathan Nguyen	Engineer	10%												
		Team Member	Stone LeCroy	Engineer	10%												
		Team Member	Cruz Saucedo	Engineer	10%												

The SIPOC diagram provides a high-level overview of the **Suppliers**, **Inputs**, **Processes**, **Outputs**, and **Customers**. The primary benefit to this diagram is the identification of customers, suppliers, and while providing a visualization of a top-level view of inputs, outputs, and processes (process flow) [4]. This provides a concise diagram to easily digest where inputs come from, how they are transformed into outputs, and where the outputs are going to be. Below is the SIPOC diagram for this project. Here we identified all parts of SIPOC as well as breaking down our process into 9 individual tasks.



Measure Phase

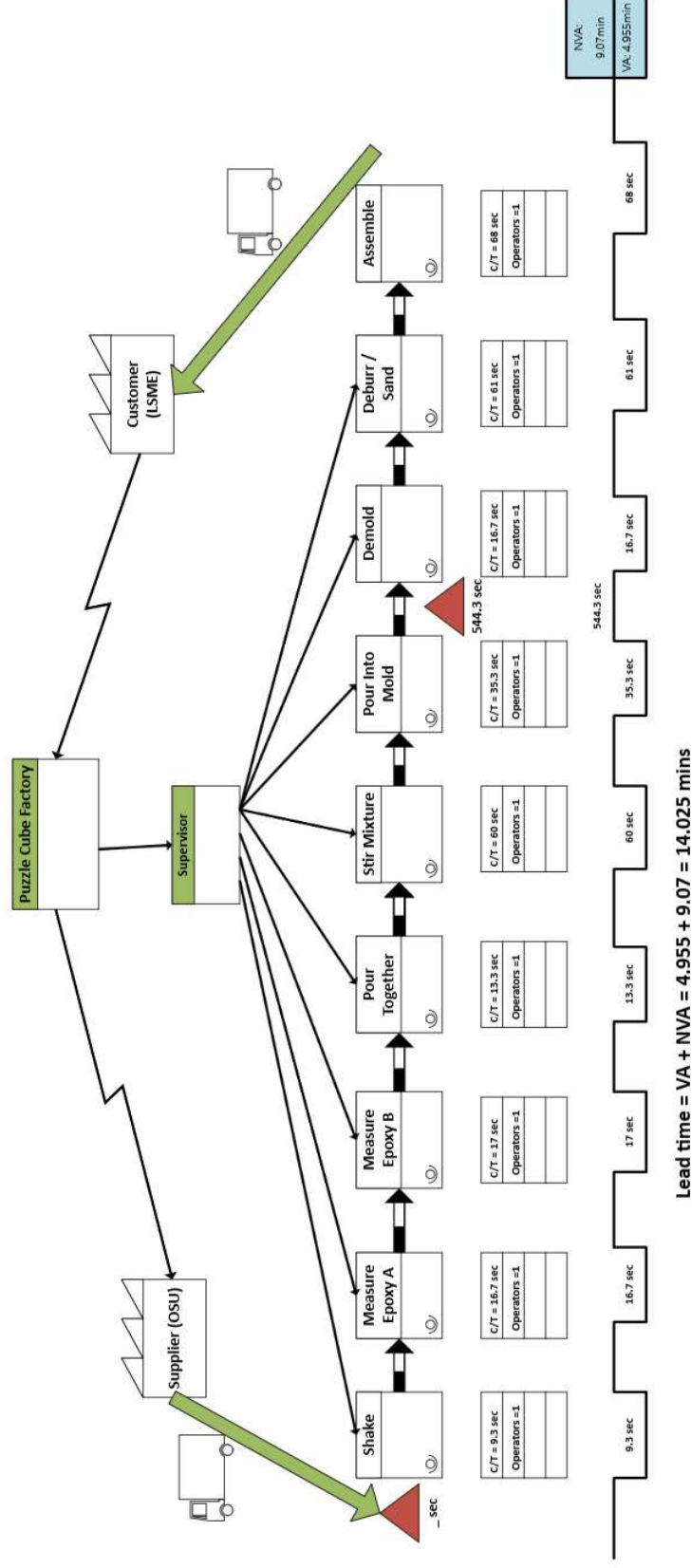
The Measure Phase of the DMAIC process involves measuring each step of the process in a quantifiable way, to be able to measure future improvements. The common phrase “what cannot be measured, cannot be improved” explains why this step is so useful. LEAN tools can be applied in the Measure phase, including a data collection plan, measurement system analysis (MSA), and value stream mapping (VSM). These tools help to visualize the data collected and organize it in a way that is easy to understand [5]. From these tools we can accurately quantify the current state of the real-world factory.

Our first step in our measurement phase was to come up with a data collection plan. Data collection plans help ensure that the data collection process is systemic, consistent, and accurate. Together, these are used to quantify a baseline of data that can guide our analysis [6]. For our time trials, we had one person perform all sequential steps while another person used a stopwatch to record how long each of the 9 different tasks took (see SIPOC for the 9 different steps). We repeated this with three different operators.

After running our time trials, we use a measurement system analysis tool (MSA) to find averages, standard deviations, minimum and maximum times, etc. (see below). From this table we can quickly see that our curing process is our longest process while our shaking of the resins is our shortest.

Operator 1					Operator 2					Operator 3				
Tool 1	Tool 2	Tool 3	R	X	Tool 1	Tool 2	Tool 3	R	X	Tool 1	Tool 2	Tool 3	R	X
1	10	shake	0		1	9	shake	0	9	1	9	shake	0	9
2	22	measure a	0	22	2	18	measure a	0	18	2	10	measure a	0	10
3	14	measure b	0	14	3	17	measure b	0	17	3	20	measure b	0	20
4	12	pour together	0	12	4	20	pour together	0	20	4	8	pour together	0	8
5	60	stir	0	60	5	60	stir	0	60	5	60	stir	0	60
6	31	pour in mold	0	31	6	50	pour in mold	0	50	6	25	pour in mold	0	25
7	992	Cure	0	992	7	600	Cure	0	600	7	441	Cure	0	441
8	19	demold	0	19	8	13	demold	0	13	8	18	demold	0	18
9	60	sand	0	60	9	74	sand	0	74	9	49	sand	0	49
10	80	assemble	0	80	10	80	assemble	0	80	10	34	assemble	0	34
11					11					11				
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15					15					15				
Mean					Mean					Mean				
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For our measuring phase, we also use a value stream map or VSM (see image below) which can be used to visually document each step and provide important information consisting of cycle time (which we have just collected), changeovertime, reliability, and operator shifts. From this we can see that our lead time is 14.02 minutes, which is behind our required takt time of .75 minutes to reach the output goal. From the VSM we can also start to see possible operation groupings, or work cells, for future production improvement.



Analyze Phase

The Analysis Phase uses the data from the Measure Phase to determine the root cause of the problem at hand. Tools such as the Fishbone diagram, “5 whys” root cause tables, and Pareto charts can help to narrow down a problem to its direct cause rather than the observed “symptoms.” Waste analysis can also be applied during this phase to identify unnecessary or avoidable waste within the process of interest [7]. Tools such as a Pareto chart can be used to identify the most significant data points based on frequency. This can be an important tool when deciding which areas of a process to improve, as it can highlight the frequency of a problem rather than focusing only on its severity or perceived severity.

The 8 wastes check sheet below helped our team identify areas with the greatest potential for improvement, based on their importance to the operation of our production system. Our team was quick to identify the defects as waste, but the check sheet helped us to also acknowledge that there was room for improvement in the motion and waiting categories as well.

8 Wastes Check Sheet					
Process Area:			Date:		
	Waste	Definition	Examples	Type "High", "Medium", "Low"	Description of Issues
D	Defects	Information, products and services that are incomplete or inaccurate	- Inaccurate applications - Broken parts - Missed deadlines	High	<i>Parts can be over or under poured creating the need for more post processing</i>
O	Overproduction	Making more of something - making it earlier or faster - than it's needed	- Extra copies of reports - Redundant storage (hard & soft) - "Reply All" on emails	Low	Over production could happen however at current capacity we are unable to produce enough
W	Waiting	Waiting for information, equipment, materials, parts or people	- Waiting for approvals - Waiting for equipment - Waiting for large batches	High	There is a lot of waiting time will waiting for the molds to cure
N	Non-Utilized Talent	Not properly utilizing people's experience, skills, knowledge or creativity	- Employees unable to make decisions - Employees not fully trained - Skilled employees doing unskilled tasks	Low	One team member could be in a job that they are not suited for, Our team is pretty willing to move people where ever they need to go
T	Transportation	Unnecessary movement of materials, information or equipment	- Hand-offs between functions - Multiple reviews - Sending, resending emails	Low	Moving components of the process can take time, having all of production at the same table helps to mitigate.
I	Inventory	Accumulation of parts, information, applications, etc. beyond what is required by the customer	- Stockpiling supplies - Information piling up for data entry - Keeping data longer than necessary	Low	Running out of materials in the middle of production can cost time and money, in this case it would be running out of mold, resin, mixing containers, and PPE
M	Motion	Any movement by people that is not of value to the customer	- Repetitive key strokes - Walking between equipment - Switching applications	Medium	Unergonomic system to measure epoxy volumes, having to bend over/ hold cup up to eye
E	Extra-Processing	Any steps that do not add value in the eyes of the customer	- Extra formatting, extra fields - Extra features, excess detail - Extra report information	Low	Cleaning of molds and work area cost time but does not add value, over stirring the mixture of A and B

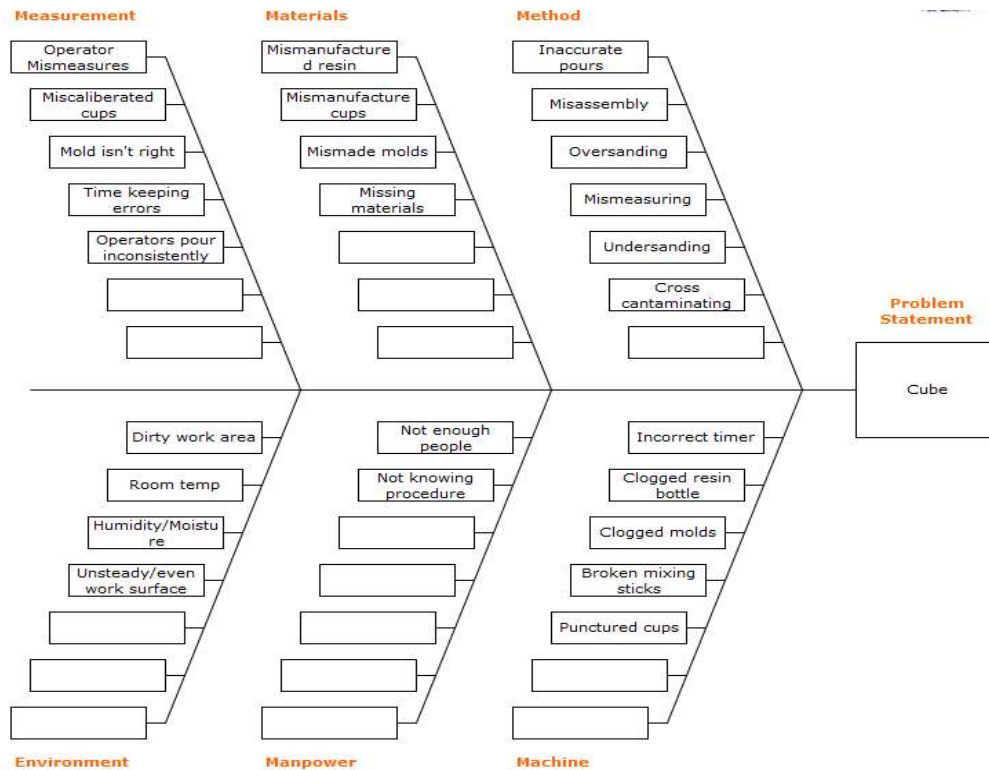
The 5 Whys and root cause analysis helped the team identify specific causes of problems that could then be solved. This helped us to realize that we needed to use more than the two molds originally provided, needed better measurements for the epoxy, and should avoid over-processing the demolded puzzles.

Title:	Root Causes:
Over or under-pouring	The cups aren't clearly marked for the exact volume needed to fill the molds
Ergonomics	The cup must be on a level surface to read it accurately, and at eye level with the operator
Cure time	It is happening because the team only had access to two molds and couldn't stage the curing process
Extra mixing	It is happening because the stated mixing time may not be the actual required time.
Extra sanding	There is no gauge to test if each part is the right size
Talent	The cells haven't been organized into specific stations

The Fishbone Diagram supported our team's methodical discovery of possible defect causes in relation to the Cube manufacturing process, using categorization of potential identified causes into several categories, as follows: Measurement, Materials, Method, Environment, Manpower, Machine.

This format allowed our team to visualize different ways that multiple influences could affect the problem, as opposed to thinking about the cause as a single source. ASQ [8] reports that the Fishbone Diagram supports structured brainstorming and is an effective way for a team to identify and examine cause/effect relationships.

We concluded during our causal analyses that there were relationships between most of the identified causes. Improper measurement, inconsistent processes, and humidity were all key components in our analysis; therefore, it is likely that multiple process variations created the Cube defects instead of a single isolated defect source.



The RPNs highlighted the fact that we thought having a quick and accurate way to measure the two parts of the epoxy and having multiple molds, allowing for the curing process to be staggered, were the most important.

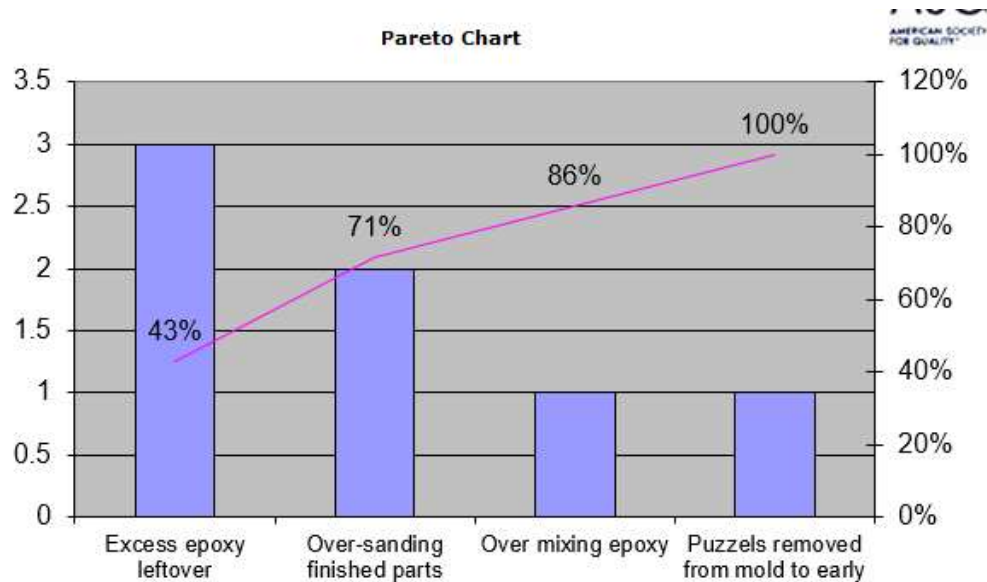
Ross:	Risk Priority Number	Stone:	Risk Priority Number	Nathan:	Risk Priority Number
Cups not marked correctly	120	Cups not marked correctly	60	Cups not marked correctly	288
Measuring needs to be ergonomic	36	Measuring needs to be ergonomic	10	Measuring needs to be ergonomic	128
Only two molds can cure at a time	200	Only two molds can cure at a time	100	Only two molds can cure at a time	96
Mixing time not accurate?	125	Mixing time not accurate?	50	Mixing time not accurate?	280
No gauge to test part tolerance	48	No gauge to test part tolerance	175	No gauge to test part tolerance	80
Cells haven't been organized	56	Cells haven't been organized	144	Cells haven't been organized	25

Cruz:	Risk Priority Number	Luncheng Zhang:	Risk Priority Number	Average:	Risk Priority Number
Cups not marked correctly	80	Cups not marked correctly	40	Cups not marked correctly	117.6
Measuring needs to be ergonomic	180	Measuring needs to be ergonomic	8	Measuring needs to be ergonomic	72.4
Only two molds can cure at a time	160	Only two molds can cure at a time	200	Only two molds can cure at a time	151.2
Mixing time not accurate?	60	Mixing time not accurate?	12	Mixing time not accurate?	105.4
No gauge to test part tolerance	75	No gauge to test part tolerance	150	No gauge to test part tolerance	105.6
Cells haven't been organized	35	Cells haven't been organized	42	Cells haven't been organized	60.4

The biggest reason for any larger variation in RPNs is likely due to the subjective nature of the ranking system between 1-10. Each group member also seems to have ranked some sections higher based on which station that were working on and what problems they ran into. For the most part though our team agreed on the ranking that was produced from the average RPNs.

The Pareto chart in the figure depicts a visual representation of the principal causes for defects within the Cube Production Process. According to ASQ, "A Pareto chart indicates the major contributors to a problem by displaying the frequency of occurrence for each type of defect, thereby allowing one to identify the 'vital few' that account for the majority of issues" [9].

Our team's analysis of the Pareto chart identified that there were two main contributors to the total number of defects, "Excess Epoxy Leftover" and "Over-Sanding Finished Parts", which collectively account for 70% of all defects. Our team was able to redirect our efforts towards addressing the highest frequency causes because of the findings from this analysis.



A simple win was realizing that we had access to much more than just two molds that we could use whenever we wanted to, allowing us to better plan out our process. Another simple fix was making sure to clean the molds before each use to ensure a quality puzzle and easier clean up at the end.

The major root causes that the team identified from these tools were; curing time was limited by the number of molds, measuring was affected by cups and ergonomics, and cells have not been organized for maximum efficiency. All of these root causes will be addressed in the following Improve and Control phases.

Improve Phase

The fourth phase of the DMAIC methodology is the Improve Phase. The Improve Phase is used to identify and execute solutions to address the problems identified in the previous Analyze Phase, these improvements could be process changes, new trainings, new systems (kanbans, decouplers, etc.), new technology, etc. [10]. In the Improve Phase it's important to brainstorm solutions to problems identified earlier in the process. This is done using a variety of tools, including the 5 WHYS diagram, where the root cause of problems can be identified. Another tool, the 8 Lean Waste Check Sheets, is used to identify wastes in the production process.

In the Improve Phase another crucial framework is the Single Minute Exchange of Dies, or SMED, where the focus is to reduce changeover time to under ten minutes. This is done by introducing things such as automation and classifying changeover tasks as internal or external. Internal tasks are changeover activities that require the machine to be shut down

while they take place, and external tasks do not. This classification is similar to the classification of value-added time and non-value-added time in VSMs. In both cases, time is saved by reducing activities in a certain category. VSMs focus on reduction of non-value-added time and SMED focuses on reduction of internal tasks [11].

Once the previously mentioned tools are used to brainstorm first ideas we can use tools such as a solution selection matrix to rate and measure our design change ideas against each other so we can choose the best solution to address our problems, improving ergonomics for operators, making tasks easier, line balancing tasks so that no one operator is doing more or less work then the others, etc. [12]. After selecting our idea to go forward with we can document these plan changes in Implementation plans a Kaizen report that help to document the changes made and ensure they will be executed on the floor in a repeatable and standardized way.

After using the 5 Whys Root Cause Sheet we're able to identify both our main problems and their root causes.

Problem	Root Cause
Bottle necking at certain stations	Line balancing issue causes unequal dispersing of work
Operator slowing after sustained work	Bad ergonomics making work tire operators
Bad quality final product	Bad molds, bad pouring, under post processing of pieces

5 WHYS ROOT CAUSE ANALYSIS TEMPLATE

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DEFINE THE PROBLEM	Workers are waiting a long time for the puzzle parts to cure	
WHY IS THIS A PROBLEM?	PRIMARY CAUSE Why is it happening? 1 Only one mold is being used at a time, and it takes 10 minutes to cure Why is that? 2 The current system was designed to only use one mold Why is that? 3 Production line wasn't optimized using lean techniques Why is that? 4 Why is that? 5 ROOT CAUSE NOTE: If the final "Why" has no controllable solution, return to the previous "Why."	
CORRECTIVE ACTION TO TAKE	CORRECTIVE ACTION Redesign the cell to utilize multiple molds in a decoupler, and balance work to eliminate waiting waste	PARTY RESPONSIBLE DATE ACTION TO BEGIN DATE TO COMPLETE

Here we address our root causes with potential solutions using brainstorming, research, and asking operators for their ideas on how to fix the issue (very important in Kiazen)

Root Cause	Solution
Bad ergonomics	Improve operator ergonomics by looking at each station and identifying if it has an ergo issue, then let operator suggest possible changes to their station to improve ergo (raising platforms for taller operators, creating better pouring tools for more controlled and comfortable pours, using more ergo friendly post processing tools, etc.
Line Balancing Issues	Look at time trials for each task and see how you can group different tasks so that cycle time can even out for each station
Bad molds, bad pouring, under post processing of pieces	Find and use molds that aren't degraded, create better pouring vessels for more controlled pours as well as additional training for pourers, add an operator to the post processing step as this step takes that longest and cannot be line balanced better.

8 Wastes Check Sheet						
Process Area:			Date:			
	Waste	Definition	Examples	Type "High", "Medium", "Low"	Description of Issues	Improvement Ideas
D	Defects	Information, products and services that are incomplete or inaccurate	- Inaccurate applications - Broken parts - Missed deadlines	High	Parts are too small or too large from over/under pouring into the mold	Create better measurement tools. Pour into molds more carefully and slowly
O	Overproduction	Making more of something - making it earlier or faster- than it's needed	- Extra copies of reports - Redundant storage (hard & soft) - "Reply All" on emails	Low	Too much resin is mixed and leftover when molds are full	Cut cups down to size to ensure no over measuring of resin
W	Waiting	Waiting for information, equipment, materials, parts or people	- Waiting for approvals - Waiting for equipment - Waiting for large batches	High	People are waiting for the puzzle pieces to cure	Use more molds so it batches better and there is less waiting time for operators down stream
N	Non-Utilized Talent	Not properly utilizing people's experience, skills, knowledge or creativity	- Employees unable to make decisions - Employees not fully trained - Skilled employees doing unskilled tasks	Low	Employees with skills for certain stations are working in the wrong one	Use time trial information and employee preference to place them in their best skilled areas
T	Transportation	Unnecessary movement of materials, information or equipment	- Hand-offs between functions - Multiple reviews - Sending, resending emails	Low	Epoxy refill is across the lab	Move all necessary materials within reach of their workstations
I	Inventory	Accumulation of parts, information, applications, etc. beyond what is required by the customer	- Stockpiling supplies - Information piling up for data entry - Keeping data longer than necessary	Low	Excess WIP produced for order	Use more precise measuring to ensure no overproduction. Plan production output so you know takt time
M	Motion	Any movement by people that is not of value to the customer	- Repetitive key strokes - Walking between equipment - Switching applications	Medium	Pouring epoxy quickly and accurately into the molds is difficult	Squeeze cups to create spout for more accurate pouring. Use skilled laborers for more accurate pours
E	Extra-Processing	Any steps that do not add value in the eyes of the customer	- Extra formatting, extra fields - Extra features, excess detail - Extra report information	Medium	Overmixing of A&B epoxy, over trimming of flashing on edges	Have time so overmixing doesn't occur. Create control piece so trimmers know how much to trim

As a final step we use the 5s worksheet as a double check to ensure that the workspace is clear of clutter and that operators can complete their tasks easily. While developing the process is very important to manufacturing just as important is ensuring the environment is ideal for production.



Setup Documentation Worksheet

MACHINE:		DOCUMENTOR:			
DATE:				PAGE 1 OF 1	
SEQ #	START TIME	EVENT	ELAPSED TIME	INT	EXT
1		Measure and mix resin parts A & B together	80s	✓	
2		Pour mixed resin into molds	53s	✓	
3		Wait for molds to cure	600s	✓	
4		Demold puzzle pieces	22s	✓	
5		Deburr puzzle pieces	24s	✓	
6		Assemble	30s	✓	
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
TOTAL TIME THIS PAGE:			21:00		
TOTAL TIME ALL PAGES:			21:00		

 Setup Time Reduction Observation Form						
PROJECT NAME - Puzzle Cube Manufacturing		WORKCENTER		From Part Number to Part Number		DATE
ACTIVITY DESCRIPTION	START TIME	END TIME	ELAPSED TIME	I/E	TOOLS, GUAGES, DRAWINGS, ETC. USED	COMMENTS
Measuring of resin parts A & B					Measuring cup, mixing cup, stir stick, and timer	
Pour mixture into molds					Molds, mixing cup, and/or syringe	
Cure			10 min		Timer	
Demold					Visual inpection	
Deburr					Knife, visual inspection	
Assemble					Visual inpection	

After brainstorming some preliminary ideas for improvement, we use the below solution matrix, ranking ideas off metrics such as cost, effectiveness, time consumption, etc. To help us narrow down our ideas.

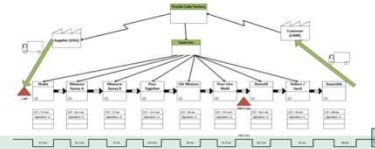
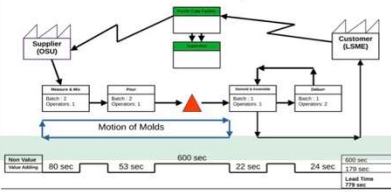
Solution Selection Matrix							
Project Goal <i>Enter Goal Statement below: (As stated on Project Charter)</i>	Please rank each solution for each criteria by using the 1-5 Scale as indicated below						
	Very Low (less good)		Moderate		Very High (best)		
	1	2	3	4	5		
Potential Solution (Provide Brief Description)	Potential to Meet Goal	Positive Customer Impact	Cost to Implement (1 = \$\$\$ & 5 = \$)	Stakeholder Buy-in	Time to Implement (1 = Long 5 = Quick)	Total Score	Implement? Yes/No
Weighted Criteria	10	9	8	7	5		
Create 4 work stations	5	3	5	4	4	165	Yes
Add a queue of curing molds between pouring and demolding	5	3	3	5	5	161	Yes
Add operator to deburr station	5	4	4	4	4	166	Yes
Cutting cups to measure size	5	5	5	3	3	171	Yes

Our implementation plan was primarily used to assign stations to operators as most of our changes were not technological improvements but rather line ordering/balancing and quick ergo improvements with cardboard boxes for workstation raising.

Streamlined Puzzle Production - Implementation Plan

Action Item (List steps required to implement solutions)	Responsible (List person(s) responsible for action steps)	Due Date (Indicate when action items must be completed)
Measure & mix resin	Nathan	Lab Day
Pour mixed resin into molds	Ross	Lab Day
Demold & Deburr demolded pieces	Lunch & Stone	Lab Day
Demold pieces & Assemble deburred pieces	Cruz	Lab Day
Cut cups down to measuring size	Nathan	Lab Day
Create timing device that notifies operators when takt time is hit	Nathan	Lab Day

Using our Kaizen Template we document the changes we intend to implement in our manufacturing system, these being primarily line balancing of tasks and ergonomic improvements.

Kaizen Report																								
Subject	Process / Project																							
Workflow																								
Before (Include pictures, diagrams, etc.)		After (Include pictures, diagrams, etc.)																						
Although we had our processes identified, we did not attempt to balance the stations to a uniform cycle time. As a result this left us with bottlenecks and waste. 	Future State Value Stream Map  <table border="1" style="width: 100%; font-size: small; margin-top: 10px;"> <tr> <td>Non Value</td> <td>80 sec</td> <td>53 sec</td> <td>600 sec</td> <td>22 sec</td> <td>24 sec</td> <td>600 sec</td> </tr> <tr> <td>Value</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Lead Time</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>279 sec</td> </tr> </table>			Non Value	80 sec	53 sec	600 sec	22 sec	24 sec	600 sec	Value							Lead Time						279 sec
Non Value	80 sec	53 sec	600 sec	22 sec	24 sec	600 sec																		
Value																								
Lead Time						279 sec																		
Benefits XQuality ___ Cost ___ Delivery X Efficiency X Waste ___ Safety ___ Energy ___ Moral ___ Other																								
More uniform cycle time between processes Reduced waste (waiting, talent, defects)																								
Originated By	Validated By	Approved By	Contact Details																					

Control Phase

The Control Phase of the DMAIC methodology is the final step in the Six Sigma improvement process and typically spans several weeks. After improvements are made and implemented throughout the Improve Phase, the goal of the Control Phase is to ensure these improvements are achieved and maintained as the production progresses. As stated by Six Sigma, this process includes developing a control plan, validating the solution, statistical analysis, implementation plans, and cost-benefit analysis. Several tools are used to ensure these happen, such as control charts, control plans, training plans, work instructions/SOPs, and financial savings reports [13].

Examining control charts further, these are common tools that are used to ensure improvements are sustainable and used as an alarm system to let you know if processes are failing. There are many different types; however, typically control charts measure the quantity of products produced over time with the addition of upper and lower bounds dictated by statistical properties. These bounds are used to verify if the variation in production overtime is within the expected range. Commonly these bounds are placed 3 standard deviations above/below the mean production [14]. This is especially applicable to a real-world setting as it allows for continuous monitoring of key performance parameters such as production. From this, we can easily see failures in the system before defects accumulate.

While control charts help identify core systemic anomalies, there are several other tools focused on the human side of production. For instance, standardized work procedures ensure standardization of tasks promoting consistency across different operator skill levels. Moreover, training plans reinforce these procedures, helping employees to better understand and adapt to the new process.

Together, these tools form a system that helps bridge the gap between improvement and long-term performance sustainability. By combining statistical monitoring and human-centered tools like training plans and SOPs, the control phase determines whether an improvement is a lasting solution or a temporary fix.

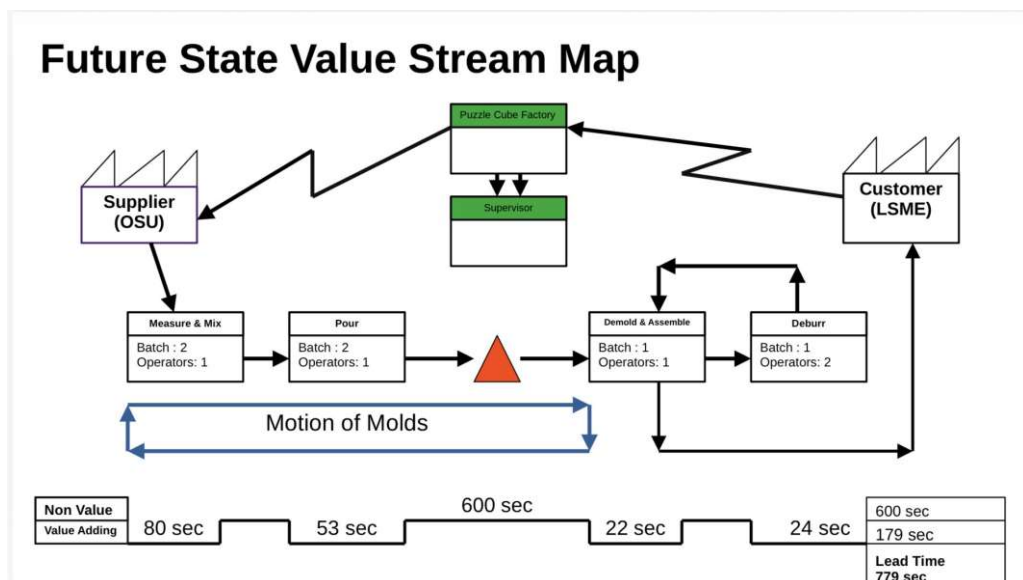
Looking at the data, there are small variations across the three trials. This is likely due to human error and slight changes in technique across all steps. Overall, the data is fairly consistent without any large failures.

Repeatability & Reproducibility of Measurement system (Gage R&R)

Operator 1						Operator 2						Operator 3						
	Trail 1	Trail 2	Trail 3	R	$\bar{X}$		Trail 1	Trail 2	Trail 3	R	$\bar{X}$		Trail 1	Trail 2	Trail 3	R	$\bar{X}$	$\bar{X}$
1	46.0			0.0	46.0	1	45.0			0.0	45.0	1	46.0			0.0	46.0	45.7
2	43.0			0.0	43.0	2	42.0			0.0	42.0	2	40.0			0.0	40.0	41.7
3	55.0			0.0	55.0	3	57.0			0.0	57.0	3	59.0			0.0	59.0	57.0
4	20.0			0.0	20.0	4	25.0			0.0	25.0	4	25.0			0.0	25.0	23.3
5						5						5						
6						6						6						
7						7						7						
8						8						8						
9						9						9						
10						10						10						
11						11						11						
12						12						12						
13						13						13						
14						14						14						
15						15						15						
Mean	41.0			-	41.00	Mean	42.3			-	42.25	Mean	42.5			-	42.50	23.3 Min $\bar{X}$
																		57.0 Max $\bar{X}$

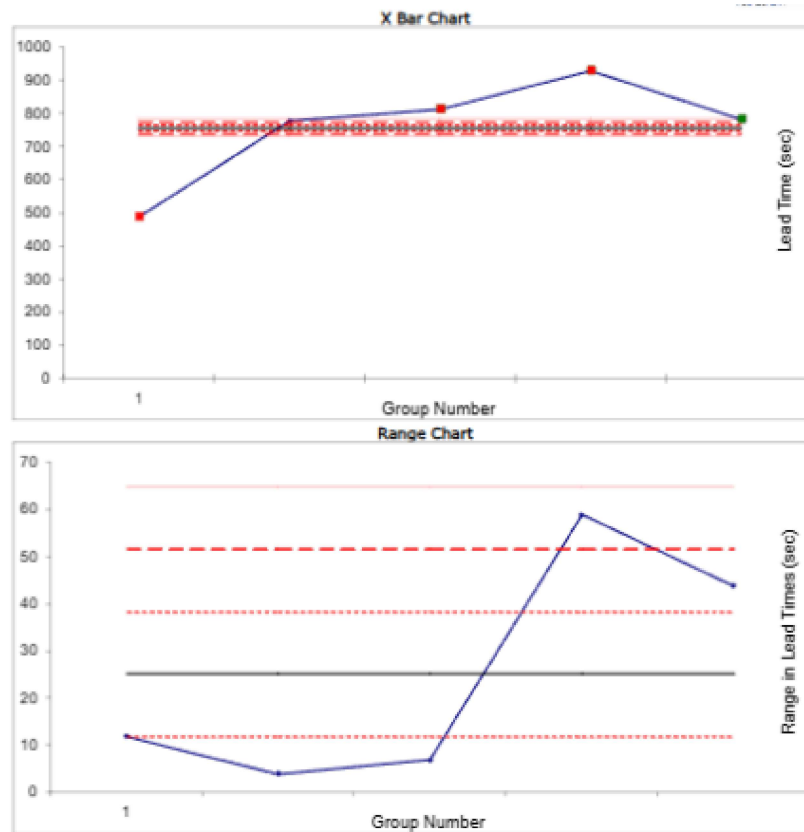
number of trials per part	1	r	$R_o = \max(\bar{X}) - \min(\bar{X}) = 1.50$ $R = -$
number of parts	4	n	
number of operators	3	k	

The primary difference between the initial and final Value Stream Map is the implementation of work cells through line balancing. Our initial VSM did not consider the worker stations; however, after using our takt time and cycle time per process we were able to use line balancing to create distinct stations. This included measuring/mixing resin, pouring, deburring, and assembling/demolding. Moreover, these changes lowered our total lead time from over 14 minutes to below 13. This is reflected in our final VSM.



The team utilized process Control Charts to gauge how in control the current process was. Since data over time was unable to be collected in our lab setting, we utilized data from a

variety of teams. In the Xbar chart, teams were unable to remain in control in their processes. Lead times across teams ranged from 500 to 900 seconds, showing that more consistent operating runs need to be recorded, and more potential changes are needed in the next iteration of the DMAIC methodology. The range chart was much more in control, within individual teams the range of lead times was much closer to being in control.



Our Standard Operating Procedures helped us specify exact quantities such as resin weight, stir time, and mixing time. Keeping track of exact quantities is crucial because it allows us to maintain consistency between mixture ratios. Without the SOP, the process would become less reliable and increase the risk of error between molds. Similarly, this outlines clear responsibilities for each station that adds structure and ensures everyone is on the same page.

STANDARD OPERATING PROCEDURE (SOP)					
SOP NO.	AUTHOR	DATE	REVISION NO.	REVISION DATE	STATUS
1	Stone LeCroy, Ross Hodson	03/11/2026			
SOP title: Puzzle Manufacturing and Assembly – Entire Process					
Process name: Puzzle Creation		Parent process:		N/A	
SOP owner: Stone LeCroy		Process time:		4.5 minutes	
Purpose of SOP: The purpose is to lay out the operating procedure for the designed puzzle manufacturing and assembly system.		Scope:		Puzzle creation from epoxy measuring to final assembly	
Definitions and abbreviations: N/A		Responsibilities (Process Stations):		Weighing And Mixing Pouring Demold and Assembly Deburring	
Process steps:		Process map / drawings:			
1. Weigh resin part A, 40 grams into one large container 2. Weigh resin part B, 35 grams extra into same container as A 3. Mix both parts of the resin for 1 minute using wooden stirring stick, pass to pouring station 4. Pour resin into 2 molds using spout fold in the large container, scrape excess epoxy over top with flat stir stick, move tray containing both molds forward into queue, replace with empty tray 5. Cure mold – 10 minutes in queue 6. De-mold both puzzles and clean mold with fingers 7. Place the unfinished puzzle in provided container 8. Pass the mold to the pouring station 9. Deburr unfinished puzzle using a razor blade 10. Assemble the deburred puzzle		<pre> graph LR A[Weighing & Mixing] --> B[Pouring] B --> C[Curing] C --> D[De-mold] D --> E[Final assembly] E --> F[Inventory/shipping] </pre>			
Related SOPs and documents:		Distribution list:			
Control Chart					
Control Plan					
Standard Work					

Figure shows an example of the Standard Work-Manufacturing template we created for our project. This tool aided our team by providing a standardized method for completing the various steps in production, with times per process, important facts and qualities for operations, and pictures serving as job aids. Standardizing the expected work sequence reduced operator to operator variation and improved process consistency. Standard work is particularly important in the Control phase since it supports the improvements made in

the earlier phases since it gives a standard for expected repeatable control of the new method. The addition of lead time, takt time, and WIP was a very handy barometer of whether this process was staying true to the improved design.

Standard Work - Manufacturing				
Company Logo		Core Process -	Date Revised:	
			Date Created:	3/11/2026
		Process Name: Process Segment:	Leadership Sponsor:	
			Process Owner:	
Lead Time	4.5min			
Takt Time	45 sec			
WIP		12		
#	Process Time	Step	Key Points & Background	Visuals & Job Aids
1	1.5	Weighing and Mixing	a. 40 gram of A, 35 grams of B, mix for 1 min	Empty paper cup passed back by the pouring station
2	0.75	Pouring and leveling the top of the mold	a. pour resin and then level off the top with a popsicle stick	Empty mold ready to be poured
3	n/a	Curing	a. 12 molds curing at different stages	n/a
4	0.375	De-mold	a. Be sure to pull any excess resin off before handing it off to be re-poured	Full mold that has been fully cured
5	1.5	De-burr	a. De-burring is done with a razor blade to clean up the rough edges leftover from the mold	Decoupler(paper cup) filled with de-molded puzzle
6	0.375	Assemble	a. Puzzles are shipped assembled and need to be assembled properly in a timely manner	Decoupler(paper cup) filled with de-burred puzzle
	4.5	TOTAL		

Figure shows our team's Six Sigma Control Plan for the puzzle creation process. This tool was helpful because it turned our improvement ideas into a plan for monitoring the daily operation. We not only identified our targets, but specified a method of control, frequency of inspection, person responsible, and plan of reaction for each step in our key processes. This forced our team to think through how we would maintain the improved performance on the process once we implemented the idea. In particular, those items we added that are quantifiable—such as quantity of materials, number of hours for curing, part quality in assembly, cycle time, and so forth—made it easier to know if we were operating in the same way as we intended. In short, the control plan connects accountability, drift control, and method of sustaining the improvements we achieved in the Improve phase.

Six Sigma Control Plan		Process: Puzzle Creation							
Champion: Ross H									
Phone:									
Black Belt: N/A									
Origination Date: 11-Mar-26									
Revision Date:									
X's	Process Step	Target	Control Method	Type	Action Required	Frequency	Responsible Person	Start Date	Reaction Plan
Epoxy measuring and pouring	Measure A	40g	Record mass	Audit	Follow Process	Every cycle	Nathan	3/11/26	Gather data on actual amount poured
	Measure B	35 g	Record Mass	Audit	Follow Process	Every cycle	Nathan	3/11/26	Gather data on actual amount poured
	Fill Molds	Slightly Overfull	Visible	Mistake Proof	Follow Process	Every cycle	Ross	3/11/26	Record over/under/exactly filled data
Cure Time	Curing	10 minutes	Time curing	Audit	Follow Process	Every cycle	Cruz	3/11/26	Gather data on time until demolding
Deburring	Deburring	Clean reduction of burrs	Record mass	Audit	Follow Process	Every cycle	Stone, Luncheng	3/11/26	Gather data on mass
Assembly	Assembly	Tightly assembled	Visible	Mistake Proof	Follow Process	Every cycle	Cruz	3/11/26	Record pass/fails
Y's	Process Step	Target	Control Method	Type	Action Required	Frequency	Responsible Person	Start Date	Reaction Plan
Cycle Time	All	< 45 sec	X bar and R chart	Control Chart	Input data into Excel	Daily	Ross	3/11/26	Ask employees on the line what went wrong, address problems creating variation

Final Simulation

All things considered the final simulation went well. Each of us knew what we needed to do and how we needed to do it to meet the demand at the end of the 4.5 minutes. At the end of the simulation, we were able to produce six complete puzzles, two of which were unfortunately rejected for quality reasons. One puzzle was rejected due to a bad mold that had gone unnoticed during set up. The second puzzle was rejected due to one of the pieces being slightly underfilled. It was also quickly apparent that the pace that the deburring station needed to move parts through would not be sustainable for the two workers in the station.

Conclusion & Future Work

The team has now completed the Define, Measure, and Analyze phases of the DMAIC methodology. These phases allowed us to observe and document the initial process as well as define the current cycle times. Measurements of each step in the process were then analyzed to determine areas for future improvement. In the next phases of the DMAIC methodology, the team will implement improvements to the areas with the largest potential for improvement and then control those new processes.

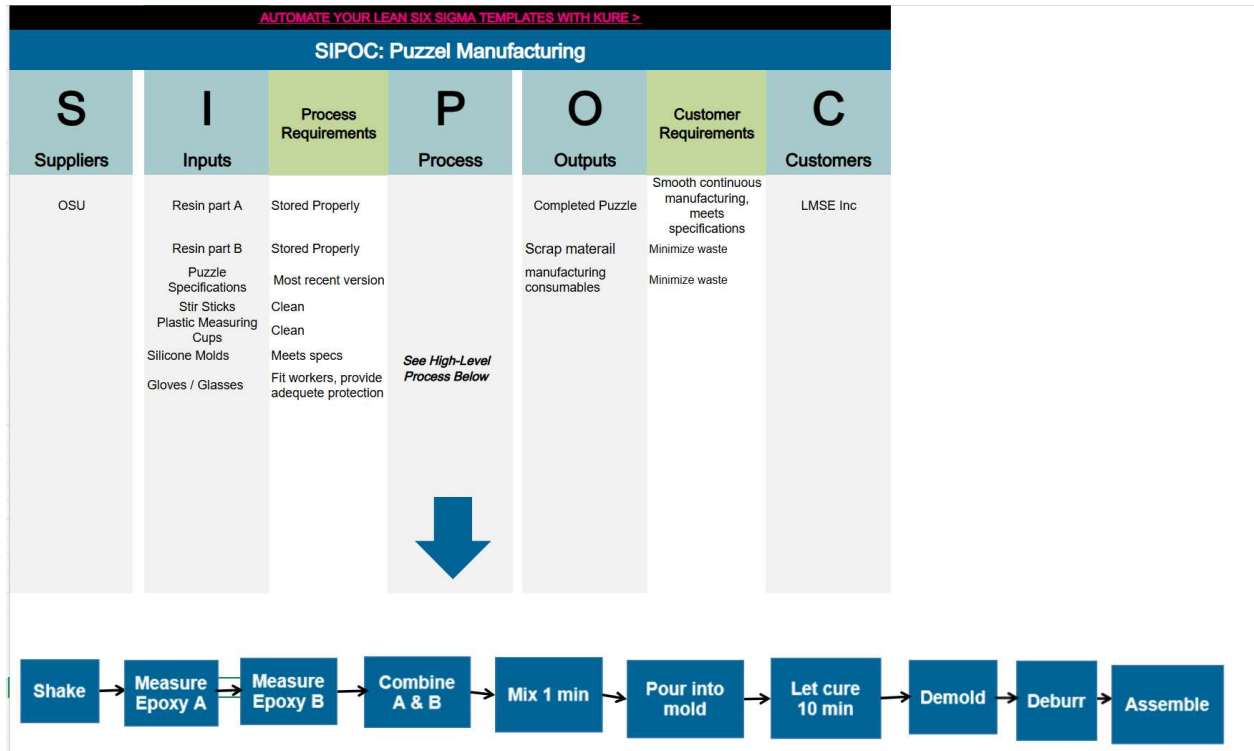
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Appendix

- SIPOC:



FMEA

Root Causes	Classification	Action Results (1-10)			
		How Severe is this factor to the process	How often does it occur?	How difficult it is to be detected?	Risk Priority Number
Ross:					
Cups not marked correctly	Tools	3	10	4	120
Measuring needs to be ergonomic	Organization	3	6	2	36
Only two molds can cure at a time	Tools	10	10	2	200
Mixing time not accurate?	Timing	5	5	5	125
No gauge to test part tolerance	Poka Yoke	4	6	2	48
Cells haven't been organized	Organization	8	7	1	56
Stone:					
Cups not marked correctly	Tools	2	10	3	60
Measuring needs to be ergonomic	Organization	2	5	1	10
Only two molds can cure at a time	Tools	10	10	1	100
Mixing time not accurate?	Timing	2	5	5	50
No gauge to test part tolerance	Poka Yoke	7	5	5	175
Cells haven't been organized	Organization	8	6	3	144
Nathan:					
Cups not marked correctly	Tools	8	6	6	288
Measuring needs to be ergonomic	Organization	4	8	4	128
Only two molds can cure at a time	Tools	6	4	4	96
Mixing time not accurate?	Timing	8	7	5	280

No gauge to test part tolerance	Poka Yoke	4	5	4	80
Cells haven't been organized	Organization	5	5	1	25
Cruz:					
Cups not marked correctly	Tools	4	10	2	80
Measuring needs to be ergonomic	Organization	6	10	3	180
Only two molds can cure at a time	Tools	8	10	2	160
Mixing time not accurate?	Timing	4	3	5	60
No gauge to test part tolerance	Poka Yoke	5	5	3	75
Cells haven't been organized	Organization	7	5	1	35
Name: Luncehng Zhang					
Cups not marked correctly	Tools	2	10	2	40
Measuring needs to be ergonomic	Organization	2	2	2	8
Only two molds can cure at a time	Tools	10	10	2	200
Mixing time not accurate?	Timing	2	3	2	12
No gauge to test part tolerance	Poka Yoke	5	10	3	150
Cells haven't been organized	Organization	6	7	1	42
Average:					
Cups not marked correctly	Tools				117.6
Measuring needs to be ergonomic	Organization				72.4
Only two molds can cure at a time	Tools				151.2
Mixing time not accurate?	Timing				105.4
No gauge to test part tolerance	Poka Yoke				105.6
Cells haven't been organized	Organization				60.4

